AZER

9:41
AZER
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AZER
Access Preference
Mobility
Hearing
Visual

Bridging gaps, empowering special
people to connect seamlessly with the world.

Product overview

Many individuals face challenges in **moving, communicating, and completing daily tasks,** which **limits their autonomy**.For those with special needs, these difficulties create reliance, not due to lack of ability, but because the world often fails to accommodate them. This project focuses on finding ways to **reduce these barriers** and enable confident, independent engagement in everyday life.

Target Audience

- **Deaf or hard-of-hearing** individuals
- **Blind or visually impaired** individuals
- People with **mobility impairments**
- Caregivers or family members (secondary audience)

Problems and Challenges

In a world built for the average, many face blurred signs, silent spaces, and inaccessible paths. **The deaf miss alarms, the blind lose direction**, and those with mobility challenges **struggle for independence**.

Technology often offers fragmented solutions. The real **need is a single, inclusive ecosystem where accessibility is built in**, where sound becomes sight, touch guides movement, and independence is universal.

Solution

We are creating an **inclusive platform** for people with hearing, visual, and mobility impairments, combining **accessibility tools** to transform sound into visuals, guide movement **through touch**, and make independence effortless.

Goal

To create a seamless, all-in-one platform that **empowers** deaf, blind, and mobility-impaired individuals to **navigate daily life independently, safely, and confidently.**

Our Approach

Discover

Identified challenges faced by deaf, blind, and mobility-impaired individuals through research, interviews, and competitor analysis.

Define

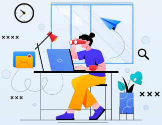
Synthesized insights into needs, goals, and problem statements, highlighting gaps in navigation, communication, and accessibility.

Ideate

Concepts focused on unified accessibility, hands-free controls, and inclusive social participation.

Design

Translated ideas into a user centric platform that enhances independence, confidence, and bridges communication and physical gaps.



User Interviews

People with hearing impairment

How do you usually communicate with people around you, including in public or important situations?

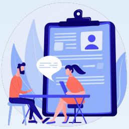
What situations or moments make communication hardest or make you feel excluded?

How has hearing loss affected your confidence or social life?

How do current apps or devices fail to meet your needs?

How do you handle alerts, alarms, or important notifications?

What features or solutions would make your daily life easier?



People with visual impairment

How do you use your phone daily with accessibility features, and which apps or tools help most?
What frustrations do you face with current accessibility apps, and which features work well or fall short?
How do you manage tasks like navigation, reading texts, identifying objects, or recognizing money using apps?
How important is offline support, real-time feedback, or combined AI and human assistance for your daily use?
How do you prefer to receive navigation cues, and what improvements would make app interfaces easier to use?

People with mobility impairment

How do you usually get around, and what situations or places are most challenging?
What devices or apps do you use, and which controls (touch, voice, gesture) work best or worst for you?
How much planning do you do before outings, and what information is hardest to find?
How often do you need help due to accessibility gaps, and how do you signal for it?
Have you ever needed emergency assistance, and what features would make you feel more confident going out alone?

Competitive Analysis

People with hearing impairment

App	Speech-to-Text & Text-to-Speech	Sign Language Translation / Learning	Real-Time & Group Communication	Offline / Low-Connectivity	AI / Smart / Emergency Features
Chat to Deaf	👍	👎	👎	👎	👎
Deaf Sign App	👎	👍	👎	👎	👎
Hand Talk Translator	👎	👍	👎	👎	👎

SignAble	👎	👍	👎	👎	👎
Deaf Communication & Mute	👍	👎	👎	👎	👎
SignMedium	👎	👍	👎	👎	👎

People with visual impairment

APP/FEATURE ---	BE MY EYES	AIRA	SEEING AI	ENVISION	TAP TAP SEE	LOOKOUT	BLINDSQUARE	KNFB Reader
HANDWRITING RECOGNITION	👎	👍	👍	👍	👎	👎	👎	👎
OCR-PRINTED TEXT READER	👍	👍	👍	👍	👎	👍	👎	👍
LIVE AI/OBJECT RECOGNITION	👍	👍	👍	👍	👍	👍	👎	👎
LIVE HUMAN ASSISTANCE	👍	👍	👎	👍	👎	👎	👎	👎
SCENE DESCRIPTION	👍	👍	👍	👍	👍	👍	👎	👎
FACE/ PERSON DESCRIPTION	👎	👍	👍	👍	👎	👎	👎	👎
NAVIGATION/ GPS GUIDANCE	👎	👍	👎	👎	👎	👍	👍	👎
BARCODE/ PRODUCT ID	👎	👎	👍	👍	👎	👍	👎	👎
CURRENCY/ MONEY DETECTION	👎	👎	👎	👎	👎	👎	👎	👎
OFFLINE MODE	👎	👎	👎	👎	👎	👎	👎	👎
BRAILLE DISPLAY SPORT	👎	👎	👎	👎	👎	👎	👎	👍
PAID	👎	👍	👎	👍	👎	👎	👍	👍

People with mobility impairment

App	Maps Accessible Locations	Wheelchair Rides	Voice / Device Control	Accessibility Reviews	Accessible Parking	Emergency / Alerts
WheelMap	👍	👎	👎	👎	👎	👎
Uber Assist / Lyft WAV	👎	👍	👎	👎	👎	👎
Google Voice Access	👎	👎	👍	👎	👎	👎
iAccess Life	👎	👎	👎	👍	👎	👎
SpotHero	👎	👎	👎	👎	👍	👎

Google Maps (Accessibility Features)	👍	👎	👎	👎	👎	👎

User Persona

Goals

- Safely and independently navigate urban/public spaces
- Access multisensory and screen-reader friendly digital content
- Avoid obstacles and reduce physical risk
- Participate confidently in group and civic life
- Promote accessibility and inclusivity

Pain Points

- Difficulty with signage, obstacles, tripping hazards
- Digital interfaces often incompatible with screen readers
- Lack of tactile/auditory cues for navigation
- Social exclusion from inaccessible content
- Emotional challenges with ongoing exclusion

Needs & Opportunities

- Reliable tactile and auditory navigation cues
- Accessible digital content and screen-reader friendly interfaces
- Real-time hazard and obstacle alerts
- Group activity features for visually impaired users
- Integrate indoor/outdoor navigation with real-time obstacle detection
- Collaborate with public spaces to improve tactile signage and guides
- Use AI for instant screen-reader optimization and hazard reporting

Neha

Age : **29**

Occupation : **Accessibility Consultant**

Location : **Hyderabad**

Disability : **Blind since childhood (mobility cane and screen reader user)**

Neha is an expert in accessibility who is blind and navigates daily life with a cane and assistive software. While skilled, she faces digital and physical barriers from poor signage, hazardous public spaces, and uncaptioned group activities..

Rohan

Age : **32**

Occupation : **Graphic Designer**

Location : **Mumbai**

Disability : **Deaf since birth**

Rohan is a visual communicator and creative professional who uses written messages and sign language. He is often frustrated with audio-centric notifications and meetings, and seeks parity across devices and daily communications.

Goals

- Receive visual/haptic alerts for alarms, calls, and emergencies
- Use real-time captions and interpreters for meetings
- Access devices via text or touch, not voice-only
- Participate fully in work and social groups
- Ensure equal access to digital and physical spaces

Pain Points

- Misses alerts due to audio-only design
- Excluded from calls and voice-based groups
- Dependence on others for translation or captions
- Delays in emergency notifications
- Frustration with tech and workplace biases

Needs & Opportunities

- Universal access to visual and haptic notifications
- Automatic live captioning and translation
- Seamless integration for visual-centric controls
- Profiles/configs for group communication accessibility
- Develop device/app systems prioritizing non-audio signals and captions
- Enable plug-and-play text/visual notifications for smart devices
- Crowdsource or AI-enable interpreter/caption services

Goals

- Seamlessly operate all devices hands-free
- Minimize pain and fatigue from physical apps and tech
- Personalize controls to her current mobility level
- Swift emergency response
- Remain empowered and productive in her job

Pain Points

- Pain from device use, fatigue after frequent actions
- Hands-free controls are fragmented across platforms
- Lacks dashboard for quick shortcut actions
- Anxiety about slow/ineffective help in emergencies

Needs & Opportunities

- Consistent, reliable hands-free device and app controls
- Customizable dashboards and quick-action features
- Automation for physical and digital tasks
- Fast, accessible emergency functions
- Centralize and standardize hands-free input options
- Adaptive dashboards tailored to user ability
- Integrate with smart home and mobility devices for independence interpreter/caption services

Nisha

Age : **36**

Occupation : **Product Designer**

Location : **Pune**

Disability : **Muscular dystrophy (uses power wheelchair, limited arm/hand mobility)**

Nisha is a talented designer working remotely, relying on a power wheelchair and adaptive technology due to limited movement. She finds touch screens and standard gestures increasingly painful, relying on voice and automation when possible, but wishes for better integrated solutions.

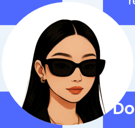
Empathy Maps

Says

- “I struggle to navigate public spaces safely without proper cues.”
- “Many digital platforms don’t work well with screen readers.”
- “I want to participate fully in group activities without barriers.”
- “Accessible design should be the standard, not an afterthought.”

Thinks

- “Urban spaces and apps often overlook the needs of visually impaired users.”
- “With better tactile, auditory, and AI tools, I could navigate independently.”
- “Accessibility is not just a personal need, it’s a societal responsibility.”



Feels

- Frustrated by poor signage, obstacles, and incompatible digital content.
- Excluded socially and professionally due to inaccessible environments.
- Concerned about physical safety while navigating urban spaces.
- Motivated to advocate for inclusive design and accessibility improvements.

Does

- Uses a cane and assistive software to navigate and access information.
- Relies on tactile and auditory feedback whenever available.
- Avoids risky or poorly designed environments when possible.
- Promotes accessibility best practices professionally and personally

Says

- “I miss important alerts because everything is audio-based.”
- “Meetings and calls are frustrating without captions.”
- “I wish devices could work without relying on voice commands.”
- “I want to participate fully in work and social settings.”

Thinks

- “Technology often doesn’t consider people like me.”
- “If I had better accessibility tools, I could be more independent.”
- “Communication should be equal across devices and environments.”

Feels

- Frustrated with tech and workplace bias.
- Excluded from group activities and calls.
- Anxious about missing alarms or emergency notifications.
- Motivated to find tools that give independence and equal access

Does

- Uses written messages and Indian Sign Language to communicate.
- Relies on others for translations or live captions in meetings.
- Adapts devices via text or touch where possible.
- Seeks solutions that provide visual/haptic alerts and real-time captioning.

Says

- “Touchscreens and standard gestures cause me pain.”
- “I wish all my devices could work seamlessly hands-free.”
- “I need quick, reliable emergency support.”
- “Customizable controls would make my work and life easier.”

Thinks

- “Current hands-free solutions are fragmented and inefficient.”
- “Automation and adaptive dashboards could reduce fatigue and improve productivity.”
- “Integrating devices with my wheelchair and smart home would give more independence.”

Feels

- Frustrated by physical strain and fatigue from device use.
- Anxious about slow or ineffective emergency responses.
- Limited by non-adaptive technology across platforms.
- Motivated to stay productive and empowered despite mobility challenges.

Does

- Uses a power wheelchair and adaptive technology for work and daily tasks.
- Relies on voice commands and automation whenever possible.
- Avoids repetitive physical interactions that cause pain.
- Seeks solutions that centralize controls, automate tasks, and provide fast emergency access.

Pain Points

Deaf or Hard of Hearing

Dependence on others for speech.
Phone/voice-only support barriers.
Missed emergencies without visuals.
Limited access to sign-language interpreters
Limited communication options can affect performance evaluations, promotions, or networking opportunities.
Feeling excluded in social gatherings where people speak without sign language support.

Blind or Low Vision

Navigation is hard without cues.
Signs/labels unreadable without audio/touch.
Unsafe stairs/elevators with poor design.
Unlabeled tech blocks use.
Need help to find places/things.

Mobility Impaired

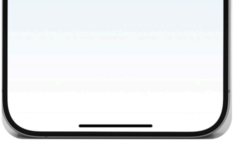
Entrances not wheelchair-friendly.
Narrow or rough walkways.
Transport not step-free.
Manual controls cause fatigue.
Restrooms/workspaces often blocked.
Poor curb cuts and uneven terrain outside.

Final Problem Statement

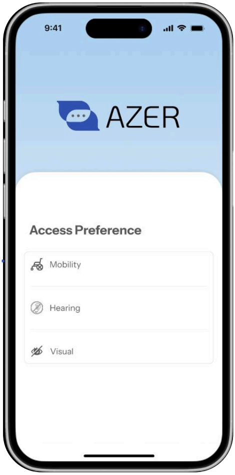
Independence should never be a dream delayed by blurred signs, silent alarms, or crowded paths. Yet for many with special abilities, everyday barriers create dependency, not from a lack of strength, but from a world that too often forgets to include them. This is a call to reimagine technology as a bridge: an app that frees movement, amplifies voices, and lights the way toward a life lived fully, fearlessly, and free.

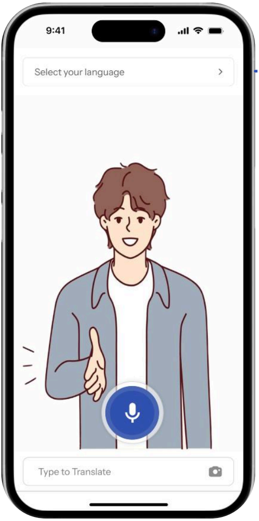
Design

The interface of AZER app

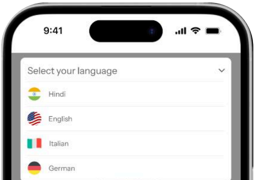


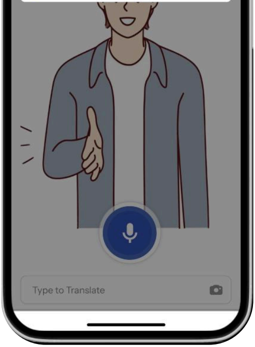
People will choose that type they need help with

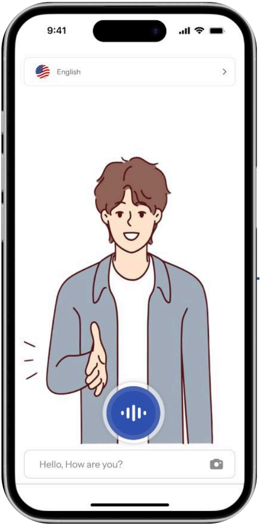

9:41
AZER
Access Preference
Mobility
Hearing
Visual


9:41
Select your language
Type to Translate

User can select their native language in which they want to communicate.

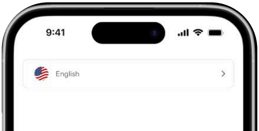

9:41
Select your language
Hindi
English
Italian
German


Type to Translate

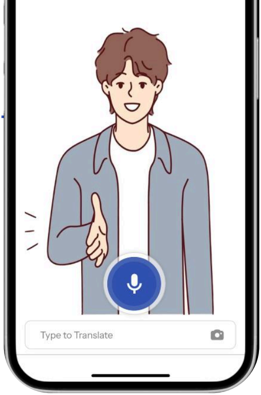

9:41
English
Hello, How are you?

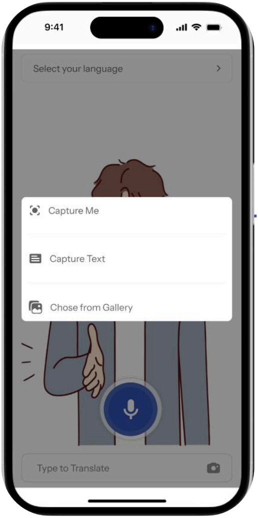
This is speech-to-text/sign language interface.

Here the speech of people the user is talking to can be translated into text as well as sign language (for those who can't read)

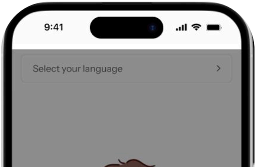

9:41
English

Text-to-speech translation is also integrated to allow user to voice his thoughts to the people he is talking to, allowing a two way communication.

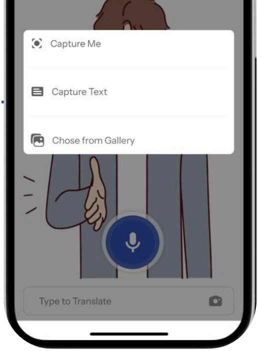

Type to Translate

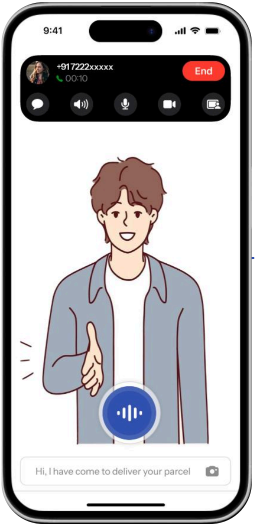

9:41
Select your language
Capture Me
Capture Text
Chose from Gallery
Type to Translate

This even allows the people who can’t text to capture their sign language and then translate it into speech.


9:41
Select your language

It also helps people who can't read to capture text which will then be changed into sign language for their understanding.


Capture Me
Capture Text
Chose from Gallery
Type to Translate


9:41
+917222xxxxx
00:10
End

Hi, I have come to deliver your parcel

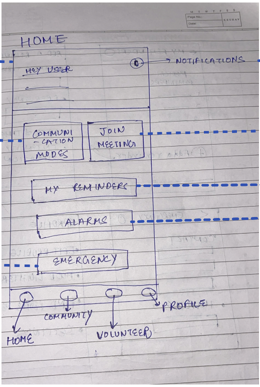
the biggest problem this platform is solving is calling by deaf or mute person. Deaf people often find it difficult to communicate through calls.
Here, the call will directly get switched to this interface where the calling be now be supported by different communication modes, allowing a smooth two way communication.

After choosing hearing
we land here.

Personalized message so as to make them feel relatable

Different communication modes like using speech-to-text, text-to-text or sign language can be done. This is the mockups that are made above.

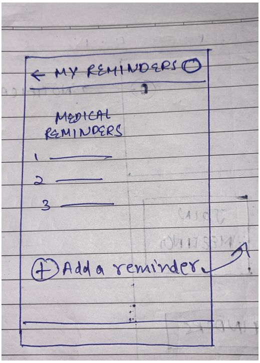
Emergency contacts can be made in case they need it.


HOME
HEY USER
NOTIFICATIONS
COMMUNI-CATION MODES
JOIN MEETING
MY REMINDERS
ALARMS
EMERGENCY
COMMUNITY
PROFILE
HOME
VOLUNTEER

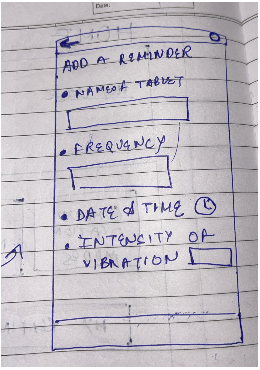
All notifications, even missed ones will be seen here.

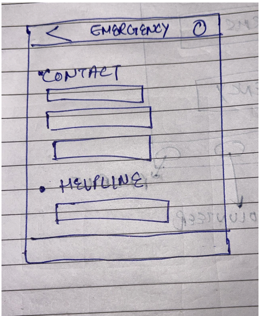
Live meetings generally does not provide good transcript, users can access meeting apps through this platform and have good experience.

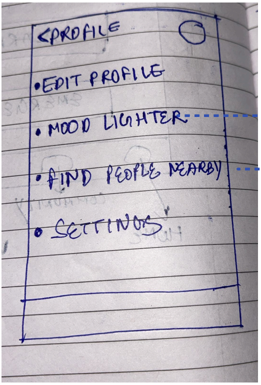
Medical reminders can be set through this. also, People will have difficulty recognizing alarms, hence a particular option can be made.


MY REMINDERS
MEDICAL REMINDERS
1
2
3
Add a reminder

Reminders can be fitted as well as seen in these interfaces.

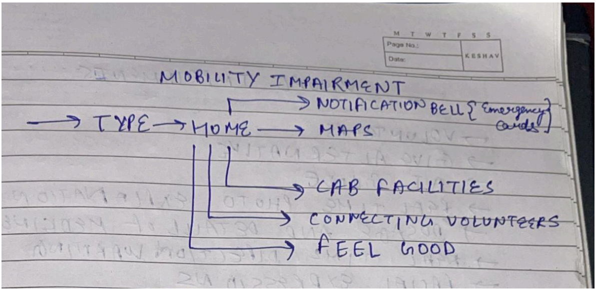

ADD A REMINDER
NAME OF TABLET
FREQUENCY
DATE & TIME
INTENSITY OF VIBRATION


EMERGENCY
CONTACT
HELPLINE

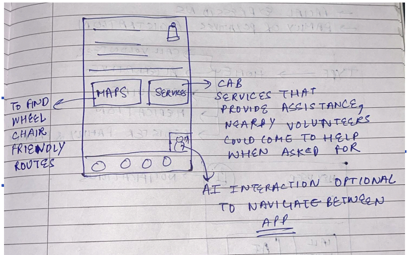

<PROFILE
•EDIT PROFILE
• MOOD LIGHTER
• FIND PEOPLE NEARBY
• SETTINGS

Mood Lighter helps user find self help or story books for the low days in their life.

Users can even connect with the people nearby, who are related to the user by the means of their special-ness.


M T W T F S S
Page No.:
Date:
KESHAV
MOBILITY IMPAIRMENT
TYPE
HOME
NOTIFICATION BELL { Emergency cards }
MAPS
CAB FACILITIES
CONNECTING VOLUNTEERS
FEEL GOOD

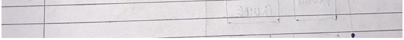
After choosing mobility
we land here.

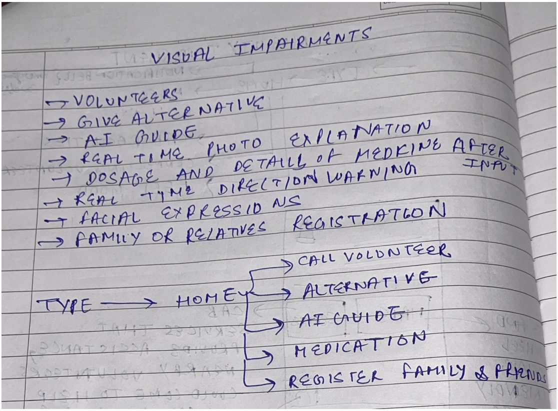

MAPS
SERVICES
TO FIND WHEEL CHAIR FRIENDLY ROUTES
CAB SERVICES THAT PROVIDE ASSISTANCE,
NEARBY VOLUNTEERS COULD COME TO HELP WHEN ASKED FOR
AI INTERACTION OPTIONAL TO NAVIGATE BETWEEN APP

Maps feature to provide the best route

services of calling a volunteer or cab to arrive the destination

AI integrated optional voice assistant to help us navigate between the app



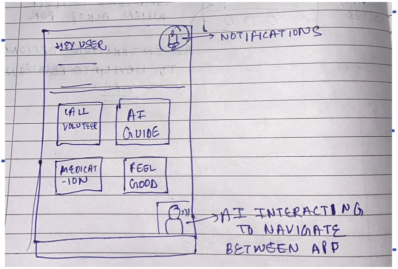


VISUAL IMPAIRMENTS

→ VOLUNTEERS
→ GIVE ALTERNATIVE
→ AI GUIDE
→ REAL TIME PHOTO EXPLANATION
→ DOSAGE AND DETAILS OF MEDICINE AFTER INPUT
→ REAL TYME DIRECTION WARNING
→ FACIAL EXPRESSIONS
→ FAMILY OR RELATIVES REGISTRATION

TYPE → HOME
→ CALL VOLUNTEER
→ ALTERNATIVE
→ AI GUIDE
→ MEDICATION
→ REGISTER FAMILY & FRIENDS

After choosing visual we land here.


HEY USER
NOTIFICATIONS
CALL VOLUTEER
AI GUIDE
MEDICAT-ION
FEEL GOOD
AI INTERACTING TO NAVIGATE BETWEEN APP

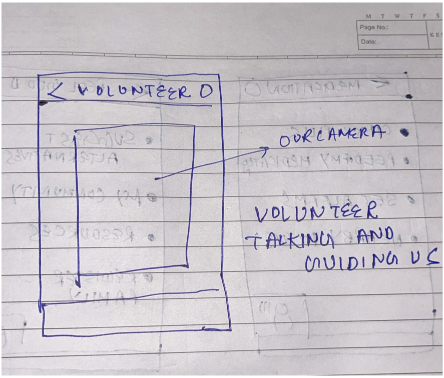
calling volunteer to help you navigate with help of your camera

medication feature for pill check via camera, setting alarms, nearby doctors.

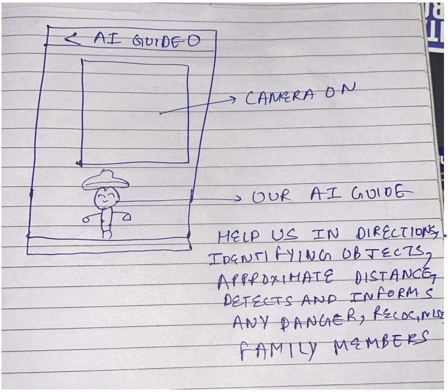
AI guide to help recognise members, directions, detecting obstacles etc.

feel good feature to suggest alternatives, connect with community etc

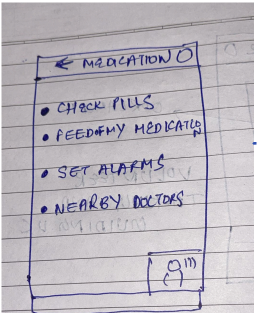
AI integrated voice assistant to help us navigate between the app


M T W T F S
Page No.:
Date:
< VOLUNTEER O
OUR CAMERA
VOLUNTEER TALKING AND GUIDING US

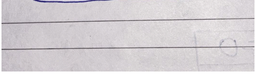
calling volunteer to help you navigate with help of your camera

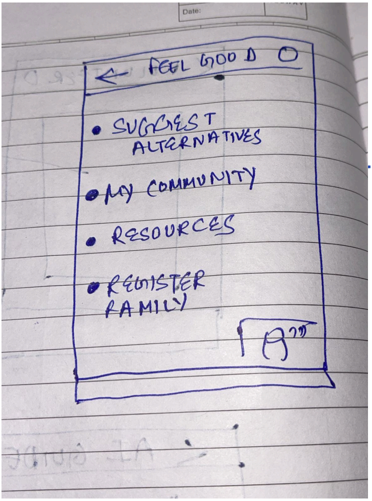

< AI GUIDE O
CAMERA ON
OUR AI GUIDE
HELP US IN DIRECTIONS, IDENTIFYING OBJECTS, APPROXIMATE DISTANCE, DETECTS AND INFORMS ANY DANGER, RECOGNIS
FAMILY MEMBERS

AI guide to help recognise members, directions, detecting obstacles etc.


← MEDICATION O
CHECK PILLS
FEED MY MEDICATION
SET ALARMS
NEARBY DOCTORS

medication feature for pill check via camera, setting alarms to remind you to take pills, nearby doctors, feed you medication pill images to help recognize easily by AI




Date:
FEEL GOOD
SUGGEST ALTERNATIVES
MY COMMUNITY
RESOURCES
REGISTER FAMILY

feel good feature to suggest alternatives for playing games like badminton, connect with your community, have resources like audio books, register family members to recognise them by AI guide

LOGO

The two curved blue forms resemble speech bubbles, a universal symbol for communication, voice, and expression. For an accessibility app (especially for those with visual, hearing, or mobility impairments), this signifies:

- Every voice matters.
- Communication without barriers.
- Inclusivity in conversation and interaction.

#E1EAFE	#A1C3FD	#499EFC	#1579CD	#0C5593	#05345E	#01162D

Color Palette

AZER

9:41
2
Azer
MON
6
Calendar
Photos
Notes
Reminders

9:41
AZER
Access Preference
Mobility
Hearing
9:41
AZER

9:41
AZER
Access Preference
Mobility
Hearing
Visual
9:41
AZER